Fiscal Year:	FY 2017	**Task Last Updated:**	FY 07/13/2017
PI Name:	Kabarowski, Janusz Ph.D.		
Project Title:	Radiation-Induced Early Changes in Gene and Protein Expression, Lipid Oxidative States, and Vascular Function Related to Atherosclerosis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	janusz@uab.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	205-996-2082
Organization Name:	University of Alabama at Birmingham		
PI Address 1:	Dept. of Microbiology		
PI Address 2:	BBRB 334		
PI Web Page:			
City:	Birmingham	**State:**	AL
Zip Code:	35294	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	01/01/2017	**End Date:**	07/31/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	January 2017: New Principal Investigator is Dr. Janusz Kabarowski; Previous PI was Dr. Dennis Kucik.		
COI Name (Institution):	White, Charles Roger Ph.D. (University of Alabama, Birmingham)		
Grant/Contract No.:	NNX14AH88G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continues "Radiation-Induced Early Changes in Gene and Protein Expression, Lipid Oxidative States, and Vascular Function Related to Atherosclerosis" with Principal Investigator (PI) Dr. Dennis Kucik. Dr. Janusz Kabarowski is new PI as of January 1, 2017, due to Dr. Kucik's retirement. Epidemiological studies have established that radiation from a number of terrestrial sources increases the risk of atherosclerosis, even at relatively low doses. Although the heavy ions found in galactic cosmic radiation (GCR) interact very differently with tissues than do the high- energy photons that comprise most terrestrial radiation, it has been thought that GCR could pose a risk as well. In our previous project, we showed that heavy ions such as 0Fe and 28Si are indeed atherogenic, accelerating the development of plaques in the apoE -/- mouse model. Mouse models, however, are known (from comparison of animal experiments to human epidemiological data) to require much higher radiation doses for plaque formation than humans. This is likely due to species differences in lipid metabolism and the immune system. Plaque formation is a very late endpoint, however. While species differences may lead to divergence in pathological consequences at times distant from absorption of radiation, very early steps, such as changes in gene expression, protein expression, and lipid oxidation, are likely to be similar in humans and mice. Thus, analysis of these early changes in response to space-relevant doses of 56Fe and protons will enable us to better predict risk of cardiovascular consequences from space radiation and increase our understanding of the mechanisms involved. In addition, a focus on these early events will obviate the need for models incorporating unnaturally high plasma lipid and cholesterol levels, so that healthy astronauts can be more closely simulated. Our hypothesis is that irradiation of blood vessels in vivo leads to early species-independent changes in endothelial cell genes, proteins and lipids that elicit vascular dysfunction and are therefore likely to be predictive of cardiovascular disease. An understanding of these changes can be used to help predict risk and propose countermeasures. We will test this hypothesis through completion of the following specific aims: Aim 1: Determine, as a function of radiation dose, acute, radiation-induced atherosclerosisassociated changes in gene and protein expression in aortic endothelial cells of irradiated mice. Aim 2: Determine, as a function of radiation dose, changes in oxidation states of aortic endothelial cell lipids. Aim 3: Determine the dose dependence of functional consequences of 56Fe, and protons with respect to perturbations in endothelial reactivity and aortic adhesivity. Aim 4: Develop a systems biology model of response of aortic endothelial cells to radiation damage to determine molecular mechanisms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Atherosclerosis is a major adverse effect of therapeutic radiation. It is not always possible to avoid exposure of the heart and major arteries, especially for treatment of head-and-neck cancers and breast cancer. This project will provide important new information about the mechanism of radiation-induced atherosclerosis to help in designing possible mitigating strategies. Moreover, the use of ion beams for cancer treatment is becoming ever more prevalent. In the U.S., proton therapy is becoming more widespread, and carbon-ion therapy is available in Europe and Asia. Little is known about the cardiovascular effects of these new radiation treatment modalities. This work will greatly enhance our understanding of possible adverse effects.
Task Progress:	We have arranged our results from this, our third year of funding (Ed. note 7/13/2017: previous years' funding was with Dr. Dennis Kucik as Principal Investigator), in terms of progress on Roadmap questions below. Degen-1, “How can tissue specific risk models be developed for the major degenerative tissue risks, including heart, circulatory, endocrine, digestive, lens and other tissue systems in order to estimate GCR and SPE (solar particle events) risks for degenerative diseases?” Using an ex vivo experimental system for aortic monocyte adhesion that we developed, we determined last year that increased vascular adhesion is a major early event following exposure to space relevant doses of 56Fe radiation. We have now determined by pertussis toxin incubation of monocytes prior to addition to ex vivo aortic preparations from control and 56Fe irradiated C57Bl/6J mice that this adhesive effect requires signaling from Gi class of G proteins to which chemokine receptors typically signal to mediate firm adhesion of leukocytes to activated endothelium. Using a cytokine/chemokine array, we were also able to identify chemokines that are elevated in the blood of mice after exposure to 56Fe. These may prove to be useful biomarkers for cardiovascular risk prediction. Degen-2, “What are the mechanisms of degenerative tissues risks in the heart, circulatory, endocrine, digestive, lens, and other tissue systems? What surrogate endpoints do they suggest?” To understand how early vascular adhesive effects of 56Fe radiation are mediated in the vascular tissue and to identify biomarkers for better risk prediction and countermeasure development, we have completed analyses of lipids in the aorta and blood of mice 6 weeks after exposure to space relevant doses of 56Fe (a time-point at which aortic adhesion is maximal). The numbers of lipids changed in the aortae by 56Fe exposure of the mice depended on the dose of radiation and several important lipid classes already linked to vascular inflammation and/or atherosclerosis were elevated, including sphingomyelins that are known to have inflammatory effects on smooth muscle cells and/or other vascular cell types, and ceramides that may be associated with vascular cell death. Most interestingly in terms of possible countermeasures to protect against oxidation related injury caused by space radiation exposure, ether-linked phosphatidylcholines were elevated within 6 weeks of 56Fe exposure; these lipids are thought to have antioxidant properties and are thus excellent candidates for countermeasure development in the future. These results constitute progress on Aim #2: Determine, as a function of radiation dose, changes in oxidation states of aortic endothelial cell lipids, and demonstrates that 56Fe radiation causes changes in vascular and blood lipids that may predict atherosclerotic disease progression.
Bibliography Type:	Description: (Last Updated: 07/12/2017)

Abstracts for Journals and Proceedings	Kabarowski JH, Walters KB, Barnes KS, Wilson L, Gupta K, Kucik DF. "Space relevant doses of heavy ion radiation alter the lipid composition of mouse aortae, with potential consequences for radiation-induced atherosclerosis." 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017